choosing filters

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packages required

This analysis depends on the following packages:

data prep

the dataset is loaded and curred. We remove adult = 0 and NAs

```
data250_apisum_00 <- read.csv("../../../../../../01.data/2023_screen_specifics_251genotypes/0
  mutate(ivip = as.factor(ivip)) %>%
  mutate(session = as.factor(session))



data250_apisum_01 <- data250_apisum_00 %>% filter(!is.na(larvae_total_number))
data250_apisum_01$ivip <- droplevels(data250_apisum_01$ivip)

data250_apisum_02<- data250_apisum_01 %>% filter(adult_alive != "0") #remove if no adult
data250_apisum_02$ivip <- droplevels(data250_apisum_02$ivip)

data250_apisum_02$ivip <-as.factor(data250_apisum_02$ivip)
```

Once we have removed the NAs and all rows for which we don't have an adult, we can still make adjustments to the dataset. We can remove outliers to ensure that our distribution approaches a normal distribution, and we can also filter out genotypes based on a threshold for the minimum number of repetitions required to keep a genotype. We have 3 methods to handle outliers: 1)keep the outliers; 2)remove manually the outliers based on biological reasons; 3) use the interquartile range method (IQR) to remove all data outside a specific range.

The goal of this script is to study how our model reacts to changes in the outlier removal method and the minimum number of repetitions required to retain a genotype.

in the folowing loop we varry two parameters, i the minimal number of reps for the genotypes and the outlier filtering method.

function to calculate all metric used to compare filters

```
analyze_filtered_data <- function(data, i, filter_name) {
  data <- data %>%
    group_by(ivip) %>%
    filter(n() >= i) %>%
    ungroup() %>%
    droplevels()

  if (nrow(data) < 5 || nlevels(data$ivip) < 2) {
    return(NULL)  # skip iterations with too little data
  }

  NGEN <- nlevels(data$ivip)
  n_after <- nrow(data)

  # Fit the model
  mod <- lmer(larvae_total_number ~ ivip + (1|session/block), data = data)
  residuals_values <- residuals(mod)
  RMSE <- sqrt(mean(residuals_values^2))
  shapiro_res <- shapiro.test(residuals_values)$p.value
  shapiro <- shapiro.test(data$larvae_total_number)$p.value

  # Heritability model
  mod_h2 <- lmer(larvae_total_number ~ (1 | ivip) + (1 | session), data = data)
  estimates <- as.data.frame(VarCorr(mod_h2))
  genetic_variance <- estimates$vcov[1]
  residual_variance <- estimates$vcov[3]
  nbE <- nlevels(data$session)
  H2 <- genetic_variance / (genetic_variance + (residual_variance / nbE))
```

```
  return(data.frame(
    filter = filter_name, i = i, RMSE = RMSE, n_after = n_after,
    H2 = H2, shapiro = shapiro, shapiro_res = shapiro_res, NGEN = NGEN
  ))
}
```

analyse of the filtered datasets

```
# Initialize an empty data frame to store results
results <- data.frame(x = numeric(0), i = numeric(0), RMSE = numeric(0),
                      n_after = numeric(0), H2=numeric(0),shapiro=numeric(0),shapiro_res=nu


###filter based on IQR
    # Calculate IQR and bounds for outlier removal
    IQR_value <- IQR(data250_apisum_02$larvae_total_number)
    lower_bound <- quantile(data250_apisum_02$larvae_total_number, 0.25) - 1.5 * IQR_value
    upper_bound <- quantile(data250_apisum_02$larvae_total_number, 0.75) + 1.5 * IQR_value
    filter ="IQR"
    # Filter out outliers
    data_iqr <- data250_apisum_02[data250_apisum_02$larvae_total_number >= lower_bound & data


for (i in 2:9) {
  res <- analyze_filtered_data(data_iqr, i, filter_name = "IQR")
  if (!is.null(res)) results <- rbind(results, res)
}

for (i in 2:9) {
  res <- analyze_filtered_data(data250_apisum_02, i, filter_name = "no_filter")
  if (!is.null(res)) results <- rbind(results, res)
    }

data_manual<- data250_apisum_02 %>% filter(larvae_total_number != "0")

for (i in 2:9) {
  res <- analyze_filtered_data(data_manual, i, filter_name = "Manual_filter")
  if (!is.null(res)) results <- rbind(results, res)
}
```

```
print(results)
```

```
          filter i     RMSE n_after        H2      shapiro  shapiro_res NGEN
1            IQR 2 6.657805    2260 0.4108407 5.820174e-07 3.813894e-06  243
2            IQR 3 6.666687    2248 0.4112436 6.649820e-07 4.567990e-06  237
3            IQR 4 6.690371    2215 0.4113008 7.202519e-07 6.179531e-06  226
4            IQR 5 6.682081    2191 0.4114434 1.052337e-06 4.305833e-06  220
5            IQR 6 6.684478    2176 0.4136635 1.174087e-06 3.925160e-06  217
6            IQR 7 6.691100    2128 0.4163125 1.538742e-06 9.974474e-06  209
7            IQR 8 6.695692    2051 0.4062488 2.121472e-06 2.534664e-06  198
8            IQR 9 6.696062    1851 0.4047060 3.141034e-06 1.104593e-05  173
9      no_filter 2 7.481163    2343 0.3705465 3.980241e-11 1.796814e-14  243
10     no_filter 3 7.492743    2331 0.3716888 3.336373e-11 2.262427e-14  237
11     no_filter 4 7.519257    2301 0.3714715 4.682873e-11 3.553872e-14  227
12     no_filter 5 7.516380    2273 0.3709082 3.906445e-11 3.071967e-14  220
13     no_filter 6 7.516380    2273 0.3709082 3.906445e-11 3.071967e-14  220
14     no_filter 7 7.555582    2231 0.3690504 1.384709e-10 5.371531e-14  213
15     no_filter 8 7.568168    2154 0.3636776 5.928548e-10 3.184938e-14  202
16     no_filter 9 7.612903    2018 0.3636364 9.730073e-10 8.508649e-14  185
17 Manual_filter 2 7.317314    2332 0.3832690 3.836452e-11 1.785912e-12  243
18 Manual_filter 3 7.328357    2320 0.3844173 3.176391e-11 2.268226e-12  237
19 Manual_filter 4 7.353900    2290 0.3841713 4.205613e-11 3.497580e-12  227
20 Manual_filter 5 7.350153    2262 0.3837297 3.918677e-11 2.875336e-12  220
21 Manual_filter 6 7.350153    2262 0.3837297 3.918677e-11 2.875336e-12  220
22 Manual_filter 7 7.386433    2220 0.3820155 1.447377e-10 5.260479e-12  213
23 Manual_filter 8 7.375090    2136 0.3814073 8.414128e-10 5.034994e-12  201
24 Manual_filter 9 7.437219    2008 0.3775566 1.157443e-09 1.057466e-11  185
```

Based on these results it seems appropriate to select the IQR filter as it yield best heritability, lower RMSE and more normal data.

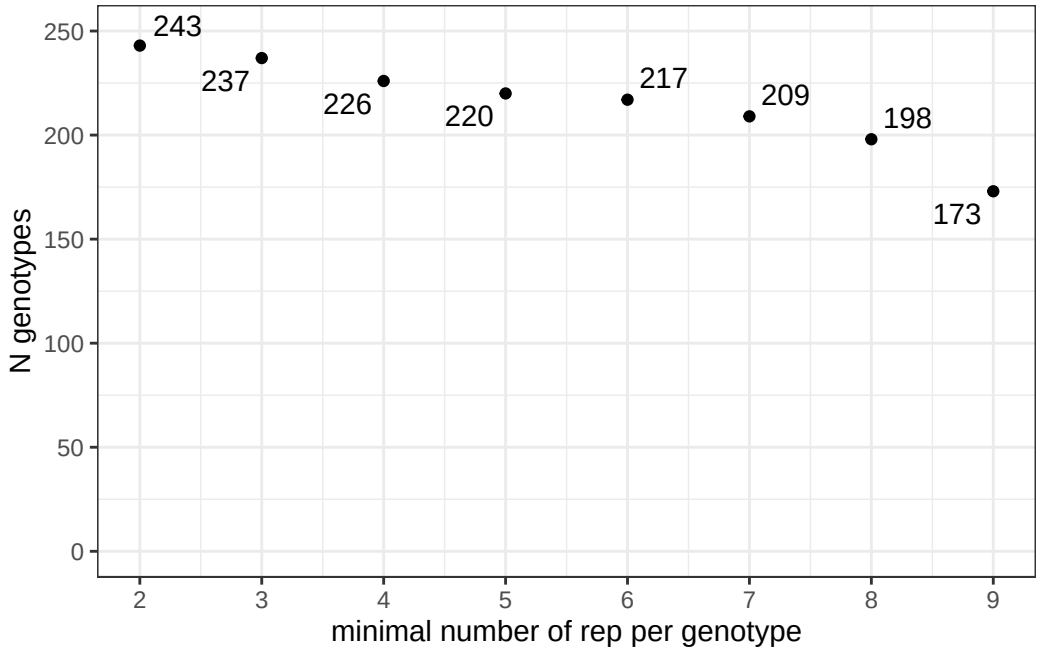
```
write.csv(x = results,
          file = "filter_annalysis.csv",
          quote = F, row.names = F)
```

```
iqr_results <- results %>% filter(filter == "IQR")
```

We now need to visualize the data we generated from the IQR filter to choose the best value for i (minimal number of repetitions to keep a genotype)

Advanced visualisation of the data generated

```
ggplot(iqr_results, aes(x = i, y = NGEN)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,250)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = NGEN), arrow = T) +
  labs(x = "minimal number of rep per genotype", y = "N genotypes")
```



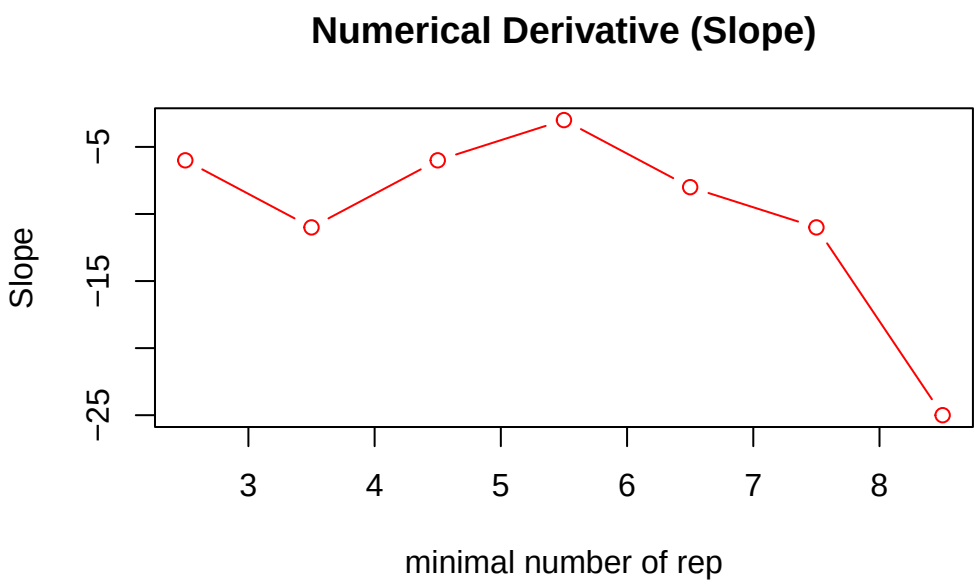

```
y<-as.vector(iqr_results$NGEN)
x<-as.vector(iqr_results$i)

# Compute slope (finite differences)
slope <- diff(y) / diff(x)

# To align slopes with x-values, use midpoints
x_mid <- (x[-1] + x[-length(x)]) / 2

# Plot slope (derivative)
plot(x_mid, slope, type = "b", col = "red", main = "Numerical Derivative (Slope)",
```

```
     ylab = "Slope", xlab = "minimal number of rep")
abline(h = 0, col = "gray", lty = 2)  # reference line for zero slope
```



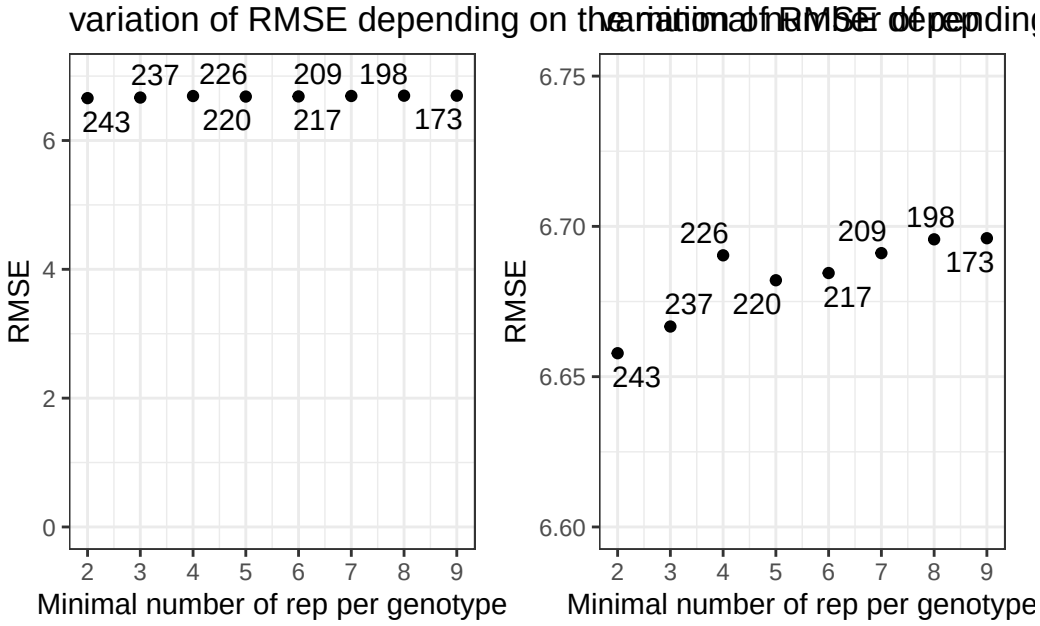
The decrease in genotype number seems quite progressive compared to A. fabae. We can still observe an increase in the genotype loss rate once once i is higher than 6. We can probably choose a value for i that is higher than the one we choose for A. fabae as we don't risk to lose too many genotypes.

```
away <- ggplot(iqr_results, aes(x = i, y = RMSE)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,7)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = NGEN), arrow = T) +
  labs(x = "Minimal number of rep per genotype", y = "RMSE") +
  ggtitle("variation of RMSE depending on the minimal number of rep")

close<- ggplot(iqr_results, aes(x = i, y = RMSE)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(6.6,6.75)) +
  theme_bw() +
```

```
  ggrepel::geom_text_repel(aes(label = NGEN), arrow = T) +
  labs(x = "Minimal number of rep per genotype", y = "RMSE") +
  ggtitle("variation of RMSE depending on the minimal number of rep")

away+close
```

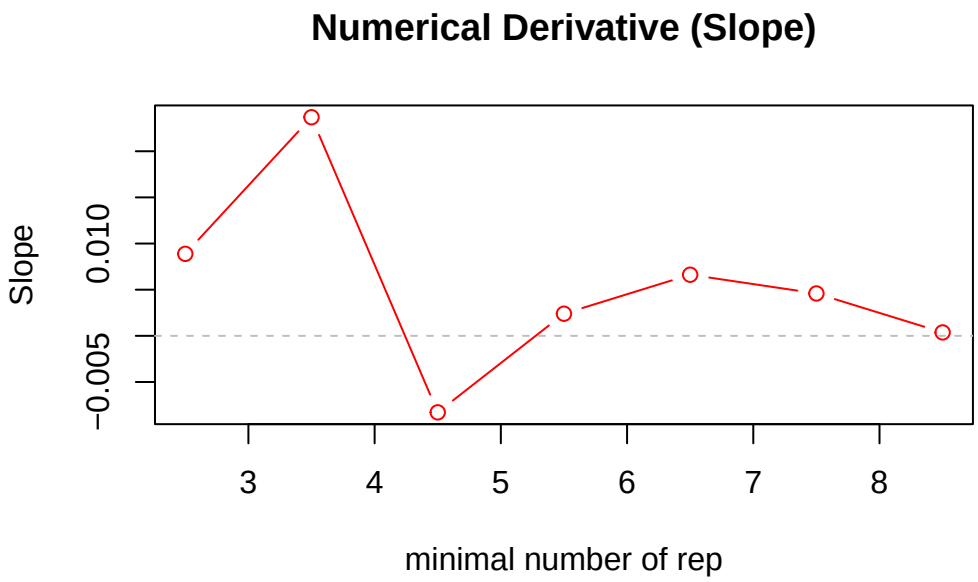



```
y<-as.vector(iqr_results$RMSE)
x<-as.vector(iqr_results$i)

# Compute slope (finite differences)
slope <- diff(y) / diff(x)

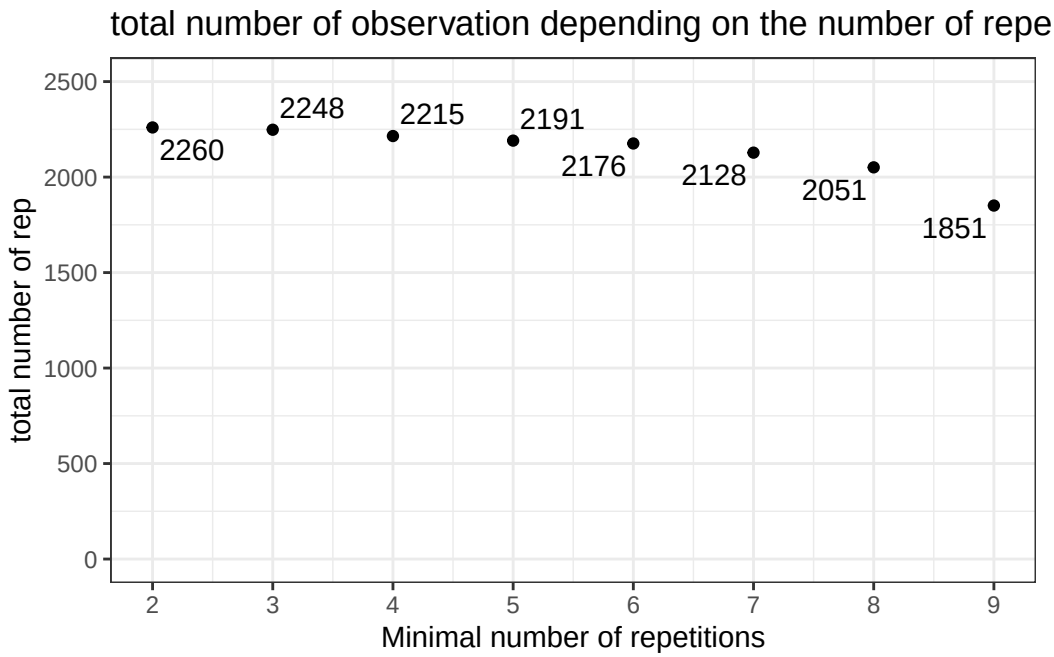
# To align slopes with x-values, use midpoints
x_mid <- (x[-1] + x[-length(x)]) / 2

# Plot slope (derivative)
plot(x_mid, slope, type = "b", col = "red", main = "Numerical Derivative (Slope)",
     ylab = "Slope", xlab = "minimal number of rep")
abline(h = 0, col = "gray", lty = 2)  # reference line for zero slope
```



It is complicated to select an optimal value for i using RMSE as the value don't vary much.

```
ggplot(iqr_results, aes(x = i, y = n_after)) +
  geom_point() +
  scale_x_continuous(breaks=seq(1,13,1)) +
  scale_y_continuous(limits = c(0,2500)) +
  theme_bw() +
  ggrepel::geom_text_repel(aes(label = n_after), arrow = T) +
  labs(x = "Minimal number of repetitions", y = "total number of rep") +
  ggtitle("total number of observation depending on the number of repetitions")
```

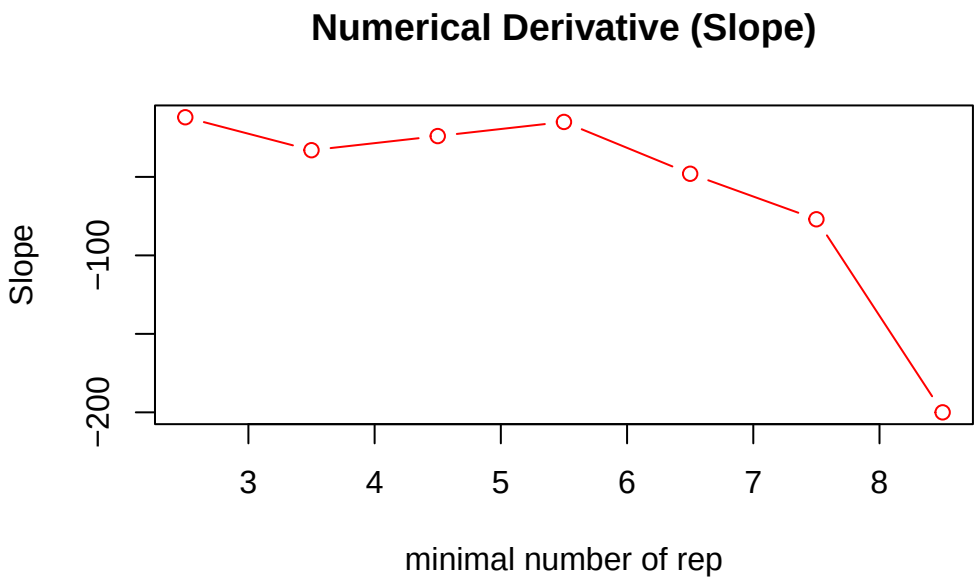



```r
y<-as.vector(iqr_results$n_after)
x<-as.vector(iqr_results$i)

# Compute slope (finite differences)
slope <- diff(y) / diff(x)

# To align slopes with x-values, use midpoints
x_mid <- (x[-1] + x[-length(x)]) / 2

# Plot slope (derivative)
plot(x_mid, slope, type = "b", col = "red", main = "Numerical Derivative (Slope)",
     ylab = "Slope", xlab = "minimal number of rep")
abline(h = 0, col = "gray", lty = 2)  # reference line for zero slope
```



The decrease in overall number of repetition is quite steady. However with the slope analysis we can see that the loss rate increase after 6 and increase again after 8

```
close<-ggplot(iqr_results, aes(x = i, y = H2)) +
  geom_point() +
  scale_x_continuous(breaks = seq(1, 13, 1)) +
  scale_y_continuous(limits = c(0.4, 0.42)) +
  theme_bw() +
  labs(
    x = "Minimal number of rep",
    y = "Heritability"
  ) +
  ggtitle("H2 depending on the number of repetitions")

away<-ggplot(iqr_results, aes(x = i, y = H2)) +
  geom_point() +
  scale_x_continuous(breaks = seq(1, 13, 1)) +
  scale_y_continuous(limits = c(0, 0.5)) +
  theme_bw() +
  labs(
    x = "Minimal number of rep",
    y = "Heritability"
  ) +
```

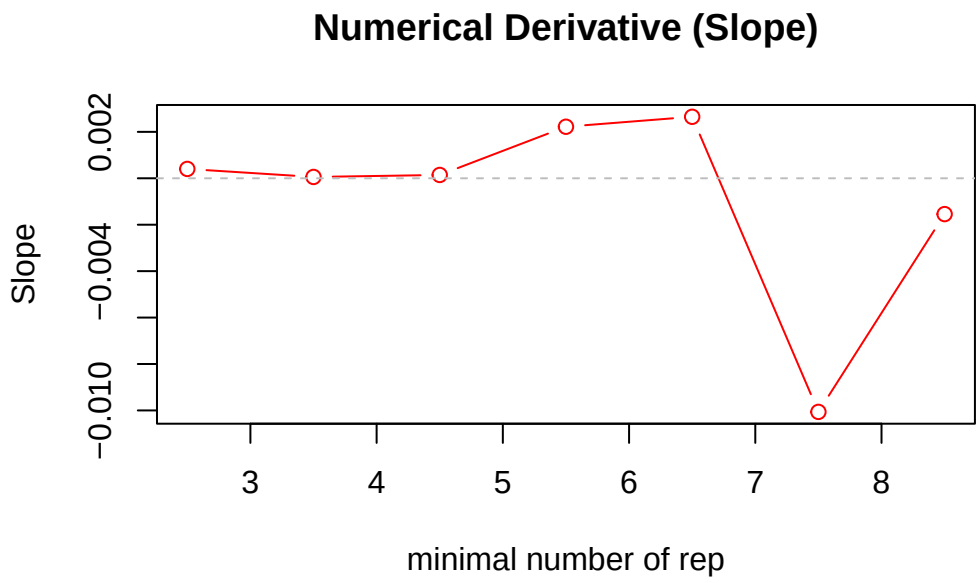
```
  ggtitle("H2 depending on the number of repetitions")

y<-as.vector(iqr_results$H2)
x<-as.vector(iqr_results$i)

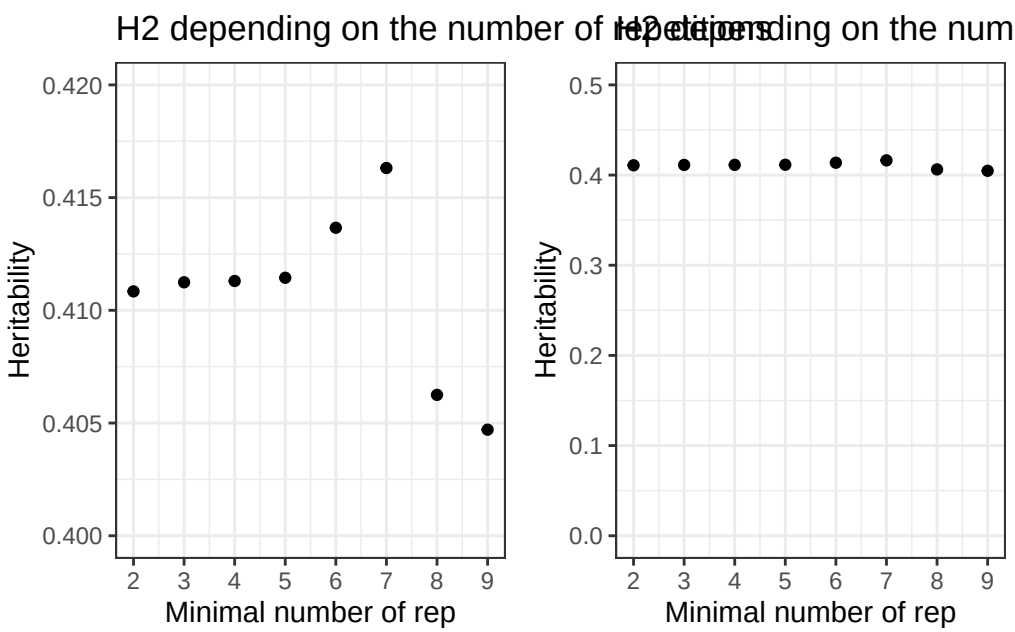
# Compute slope (finite differences)
slope <- diff(y) / diff(x)

# To align slopes with x-values, use midpoints
x_mid <- (x[-1] + x[-length(x)]) / 2

# Plot slope (derivative)
plot(x_mid, slope, type = "b", col = "red", main = "Numerical Derivative (Slope)",
     ylab = "Slope", xlab = "minimal number of rep")
abline(h = 0, col = "gray", lty = 2)  # reference line for zero slope
```




```
close + away
```



Similarly to RMSE heritability does not vary much depending on the value of i. If we zoom we can se that to maximise heritability the best value for i is 7, and the second best value is 6.

Based on all result showed here we believe that a value of 6 for the minimal number of repetition is a good fit that will allow to keep a good number of genotypes without too much noise

```
# ------------------------------
# 1. IQR filtering
# ------------------------------
IQR_value <- IQR(data250_apisum_02$larvae_total_number)
lower_bound <- quantile(data250_apisum_02$larvae_total_number, 0.25) - 1.5 * IQR_value
upper_bound <- quantile(data250_apisum_02$larvae_total_number, 0.75) + 1.5 * IQR_value

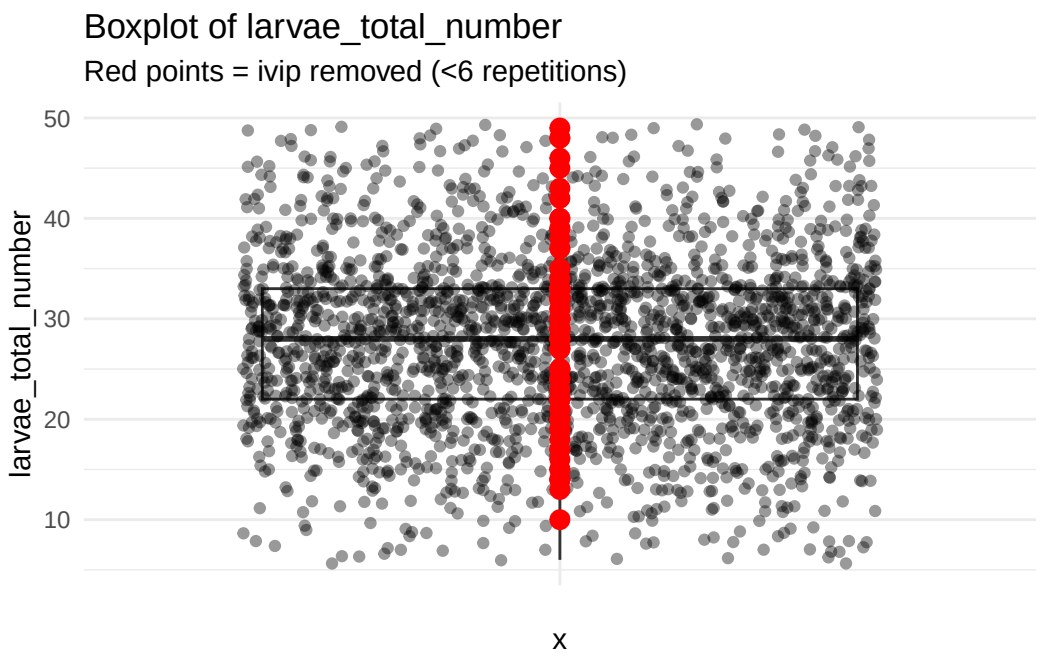
data <- data250_apisum_02 %>%
  filter(larvae_total_number >= lower_bound,
         larvae_total_number <= upper_bound)

# ------------------------------
# 2. Keep ivip with  6 repetitions
# ------------------------------
data_afabae_6 <- data %>%
  group_by(ivip) %>%
  filter(n() >= 6) %>%
```

```r
  ungroup()

# ------------------------------
# 3. Identify ivip NOT kept
# ------------------------------
ivip_not_in_6 <- setdiff(unique(data$ivip), unique(data_afabae_6$ivip))
data_removed <- data %>% filter(ivip %in% ivip_not_in_6)

# ------------------------------
# 4. Boxplot with removed ivip in red
# ------------------------------
ggplot(data, aes(x = "", y = larvae_total_number)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(alpha = 0.4) +
  geom_point(data = data_removed,
             aes(y = larvae_total_number),
             color = "red",
             size = 3) +
  labs(title = "Boxplot of larvae_total_number",
       subtitle = "Red points = ivip removed (<6 repetitions)") +
  theme_minimal()
```



```r
# ------------------------------
# 5. Table of removed ivip with genetic cluster
# ------------------------------
info_cluster <- read.csv("../../../../../../00.info/11.infos_genotypes/morphological_phenotl

ivip_removed_cluster_table <- data.frame(ivip = ivip_not_in_6) %>%
  left_join(info_cluster, by = "ivip")

ivip_removed_cluster_table
```

```
         ivip Accession.code                       Nom code.synonyme
1  IVIP100120        DVF0120 WIESELBURGER KLEINKOERNIGE
2  IVIP100199        DVF0199                HERZ FREYA
3  IVIP100227        DVF0227                    YAVNCH
4  IVIP100320        DVF0320      ULADOVSKIE FIOLETOVYE
5  IVIP100437        DVF0437                  PICARDIE
6  IVIP100633        DVF0633                   CERNAJA
7  IVIP100796        DVF0796                    AUSHRA         K1617
8  IVIP100891        DVF0891             TOSTADO CLARO
9  IVIP101050        DVF1050                   PANDOJA
10 IVIP101179        DVF1179                     MIKKO
11 IVIP101309        DVF1309                 MGD0108817
12 IVIP101414        DVF1414                  PI 430710
13 IVIP101451        DVF1451                      NEPAL
14 IVIP101497        DVF1497                 BLACK SEED
15 IVIP101508        DVF1508                       WEBO   IPK FAB6105
16 IVIP101631        DVF1631           SKLADIA KLEINE
17 IVIP101748        DVF1748                    CASTEL
18 IVIP101783        DVF1783                    CRESTA
19 IVIP101791        DVF1791                    ALFRED
20 IVIP101913        DVF1913                 PI 499959
21 IVIP101978        DVF1978               PI  223302_C
22 IVIP102080        DVF2080                     TYROL
23 IVIP102220        DVF2220                   K 0285
24 IVIP102305        DVF2305                     VF27
25 IVIP102372        DVF2372                  HIVERNA
26 IVIP102732        DVF2732                     H3025
27 IVIP103287        DVF3287                 MARIS BEAD
28 IVIP102346           <NA>                       <NA>          <NA>
   DATE.RECEPTION     Statut.biologique.SIREGAL Origine.Géographique
1        19.11.69          Traditional landrace              AUSTRIA
2        16.03.70          Traditional cultivar              GERMANY
```

3	09.04.70	Traditional landrace	ISRAEL	
4	25.04.72	Traditional landrace	UKRAINE	
5	01.03.73	Traditional landrace	FRANCE	
6	04.09.74	Traditional landrace	HUNGARY	
7	10.01.77	Traditional landrace	LITHUANIA	
8	01.09.77	Traditional landrace	PERU	
9	18.04.79	Traditional landrace	BOLIVIA	
10	11.10.81	Traditional landrace	FINLAND	
11	03.04.82	Traditional landrace	GREECE	
12	20.09.82	Traditional landrace	ALGERIA	
13	31.01.84	Traditional landrace	NEPAL	
14	01.01.80	Traditional landrace	UNKNOWN	
15	21.09.87	Advanced or improved cultivar	GERMANY	
16	01.02.90	Advanced or improved cultivar	GERMANY	
17	07.02.92	Advanced or improved cultivar	FRANCE	
18	10.02.93	Advanced or improved cultivar	UNITED KINGDOM	
19	15.10.93	Advanced or improved cultivar	NETHERLANDS	
20	01.10.93	Traditional landrace	CHINA	
21	01.10.93	Traditional landrace	AFGHANISTAN	
22	19.02.97	Advanced or improved cultivar	AUSTRIA	
23	01.06.01	Traditional landrace	CHINA	
24	01.03.05	Traditional landrace	ETHIOPIA	
25	01.03.05	Advanced or improved cultivar	GERMANY	
26	01.01.06	Traditional landrace	CHINA	
27	03.12.09	Advanced or improved cultivar	UNITED KINGDOM	
28	<NA>	<NA>	<NA>	

	HIVER.PRTPS
1	P
2	P
3	
4	P
5	P
6	H
7	P
8	P
9	P
10	P
11	1/2H
12	
13	P
14	P
15	H
16	P

```
17        1/2H
18           P
19           P
20           P
21           P
22           P
23           P
24           P
25           H
26           P
27           P
28        <NA>
                                                    INTERET..Caractères.spéciaux
1
2                                                         S rouille, autofertile
3
4
5
6
7
8
9
10                                                         petite graine précoce
11
12
13
14                                                                      PAUCIJUGA
15
16
17
18
19                                                   croissance semi determinée
20
21
22
23
24 PAUCIJUGA, buisson, lignée cartographiés par Anna Torres université de cordoue
25
26
27               R. rouille, R. bot., R. Oïdium, R. BLRV, croissance semi-déterm
28                                                                          <NA>
        ESPECE.et.sous.espèce TANNINS  SumTem
1       Vicia faba var. equina       T  978.30
```

```
2       Vicia faba var. equina       T  920.05
3        Vicia faba var. major       T  724.60
4       Vicia faba var. equina       T  995.90
5       Vicia faba var. equina       T  978.30
6       Vicia faba var. equina       T 1142.65
7        Vicia faba var. minor       T  920.05
8        Vicia faba var. major       T  847.30
9       Vicia faba var. equina       T  978.30
10       Vicia faba var. minor       T  782.45
11       Vicia faba var. major       T  884.95
12       Vicia faba var. minor       T  920.05
13       Vicia faba var. major       T  809.80
14 Vicia faba subsp. paucijuga       T  884.95
15      Vicia faba var. equina       T 1032.00
16      Vicia faba var. equina       T  920.05
17      Vicia faba var. equina       T  978.30
18      Vicia faba var. equina     ZT1  825.80
19      Vicia faba var. equina       T  920.05
20       Vicia faba var. major       T  884.95
21       Vicia faba var. minor       T  902.35
22       Vicia faba var. minor     ZT2  902.35
23      Vicia faba var. equina       T  920.05
24 Vicia faba subsp. paucijuga       T  902.35
25      Vicia faba var. equina       T 1060.50
26       Vicia faba var. major       T  748.10
27       Vicia faba var. minor       T  995.90
28                       <NA>    <NA>      NA
                       STATUT.Biologique  cluster_name
1                           trad_LANDRACE         North
2                           trad_CULTIVAR         North
3                           trad_LANDRACE         South
4                           trad_LANDRACE         North
5                           trad_LANDRACE         North
6  trad_CULTIVAR / Traditional landrace ?        Winter
7  trad_CULTIVAR / Traditional landrace ?         North
8                           trad_LANDRACE South_America
9                           trad_LANDRACE South_America
10                          trad_LANDRACE         North
11                          trad_LANDRACE         South
12                          trad_LANDRACE         North
13                          trad_LANDRACE         South
14                          trad_LANDRACE     Paucijuga
15                   ADV./IMPR. CULTIVAR        Winter
```

```
16                    ADV./IMPR. CULTIVAR         North
17                    ADV./IMPR. CULTIVAR        Winter
18                    ADV./IMPR. CULTIVAR         North
19                    ADV./IMPR. CULTIVAR         North
20                          trad_LANDRACE          East
21                          trad_LANDRACE     Paucijuga
22                    ADV./IMPR. CULTIVAR         North
23                          trad_LANDRACE          East
24                          trad_LANDRACE     Paucijuga
25                    ADV./IMPR. CULTIVAR        Winter
26                          trad_LANDRACE          East
27                    ADV./IMPR. CULTIVAR         North
28                                 <NA>          <NA>
```

```r
# ------------------------------
# 6. Cluster summary table
# ------------------------------
# Keep only non-empty, non-NA cluster_name
ivip_with_cluster <- data %>%
  distinct(ivip) %>%
  left_join(info_cluster, by = "ivip") %>%
  filter(!is.na(cluster_name), cluster_name != "")

cluster_summary <- ivip_with_cluster %>%
  group_by(cluster_name) %>%
  summarise(
    n_original = n(),
    n_removed = sum(ivip %in% ivip_not_in_6)
  ) %>%
  arrange(cluster_name)

cluster_summary
```

```
# A tibble: 6 x 3
  cluster_name  n_original n_removed
  <chr>              <int>     <int>
1 East                  24         3
2 North                102        12
3 Paucijuga              8         3
4 South                 71         3
5 South_America         18         2
6 Winter                20         4
```